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J. Experimental Child Psychology 84 (2003) 63–76

Journal of
Experimental
Child
Psychology

www.elsevier.com/locate/jecp

Time and number discrimination in a bisection task with a sequence of stimuli: A developmental approach

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Received 30 July 2001; revised 23 October 2002

Abstract

Children, aged 5 and 8 years, and adults were tested in a bisection task with a sequence of stimuli in which time and number co-varied. In a counting and a non-counting condition, they were instructed either to process the duration of this sequence while ignoring the number of stimuli (temporal bisection), or to process the number of stimuli while ignoring the duration (numerical bisection). In the temporal bisection task, number interfered with the 5-year-olds' temporal performance, indicating that young children did not process time and number independently in a sequence of stimuli when they had to attend to duration. However, number interference decreased both with age and counting strategy. In contrast, in the numerical bisection task, duration did not interfere with numerical discrimination for any age group. © 2003 Elsevier Science (USA). All rights reserved.

Keywords: Timing; Time; Number; Counting; Bisection

Timing in children has been recently reconsidered within the framework of the most completely developed model of temporal information processing, the scalar timing theory (e.g., Gibbon, 1977; Gibbon & Church, 1984), using temporal discrimination tasks initially devised for animals and human adults (i.e., the temporal bisection and generalization tasks). These temporal tasks produced orderly data in

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children as young as 3-year-olds (Droit-Volet, 2002; Droit-Volet & Wearden, 2001, 2002; Droit-Volet, Clément, & Wearden, 2001). Furthermore, young children's timing behavior conformed well to the scalar property of variance, a characteristic of timing in animals and human adults (the requirement that the standard deviation of time judgments grows as a constant fraction of the mean time judgment, when different time values are judged). On the whole, these results demonstrate that young children are able to discriminate different durations, because the clock-based system underlying time perception in animals and human adults is functional as of an early age (for a review see Droit-Volet, 2003).

However, beyond these similarities, there are also age-related changes in timing performance. Indeed, the sensitivity to duration increased with age. Among the potential sources of developmental changes in temporal sensitivity, one of the most critical is young children's difficulty in controlling the attention allocation to time (Gautier & Droit-Volet, 2002a, 2002b). The aim of the present study was thus to investigate whether young children are able to process duration when it is not the only salient dimension of an event, notably in a sequence of events in which the number of events is as salient as the duration of that sequence.

The role of selective attention in the simultaneous processing of time and number in a sequence of events has not been investigated in humans and, more particularly, in children (Mix, 1999). However, using discrimination tasks (i.e., bisection tasks), several studies have shown that animals automatically process duration and number when these two dimensions co-vary in a sequence of events (e.g., Church & Meck, 1984; Meck & Church, 1983; Meck, Church, & Gibbon, 1985; Roberts, 1997; Roberts & Mitchell, 1994). In a developmental perspective, we might suggest a developmental shift from animal-like automatic control of responses by both time and number towards a selective-attention control by either time or number. In other words, the ability to process time and number independently in a sequence of events would appear to improve with age.

To demonstrate the role of the flexibility of cognitive processing, many studies have presented verbal participants with a stimulus involving two dimensions and instructed them to attend to one dimension and to ignore the other (Logan, 1988). In this intentional condition, the processing of one dimension is judged as automatic whenever it affects participants' performance although it is irrelevant and should be ignored (Zbrogdoff & Logan, 1986). We can thus assume that, when the participants are instructed to process either the duration or number in a sequence of stimuli, interference of the irrelevant information (number or time, respectively) on performance will decrease with increasing age.

There is ample evidence from the developmental literature that resistance to interference improves during childhood (e.g., Bjorklund & Harnishfeger, 1990; Dempster, 1995; Pascual-Leone & Baillargeon, 1994; Pearson & Lane, 1991). Indeed, several studies have shown that the failure of young children in such tasks is not always due to their knowledge, or their understanding and consciousness of the rules given by the experimenter, but rather to their difficulty in attending to focal stimulus while simultaneously ignoring task distractions. This failure would be related to the pre-frontal cortex, which is involved in selective attention, and is not completely mature

until adolescence (for a review see Dempster & Brainerd, 1995). However, critical periods of maturation occur at the end of the first year and between 3 and 7 years of age (Luria, 1961, 1973). Consequently, time or number interference should be greater in children younger than 7 years old. However, a counting strategy that permits the verbal control of the measurement of time or number should reduce the interference effects of the irrelevant dimension in young children.

However, the magnitude of interference effects with age could differ as a function of the relevant information to be considered (i.e., duration or number). Indeed, in human adults, the processing of numerical quantities seems to be less attentionally demanding than that of durations longer than 1 or 2 s. According to Dehaene and Akhavein (1995), numerical quantities are processed automatically and unintentionally. In support of this idea, Tzelgov, Meyer, and Henik (1992) found that number magnitude caused interference in a task in which the participants were asked to compare the physical size of digits, although the number was irrelevant. In contrast, temporal judgments are systematically affected by concurrent non-temporal tasks, while the reverse is not true (e.g., Brown, 1997; Thomas & Weaver, 1975; Zakay, 1989). Consequently, number should interfere with temporal discrimination more than time does with numerical discrimination, especially in younger children.

The aim of the present study was therefore to investigate, in different age groups, the interference effect both of number on temporal discrimination and of time on numerical discrimination. Children, aged 5 and 8 years, and adults were tested in a bisection task, similar to that used by Meck and Church (1983), in which time and number co-varied in a sequence of stimuli. Furthermore, participants were trained and instructed to process the duration of this sequence and to ignore the number of stimuli (Experiment 1: temporal bisection), or to process the number of stimuli while ignoring the duration (Experiment 2: numerical bisection). In each experiment, bisection performance was tested with and without counting strategies.

Experiment 1: Temporal bisection

Method

Participants

The final sample consisted of 97 participants: 29 5-year-olds (15 girls and 14 boys; mean age = 5.29 years, $SD = 0.31$), 32 8-year-olds (17 girls and 15 boys; mean age = 8.05 years, $SD = 0.28$), and 36 adults (28 women and 8 men; mean age = 21.84 years, $SD = 2.72$).

Material

A PowerMacintosh computer controlled the experimental events and recorded the data with PsyScope (Cohen, MacWhinney, Flatt, & Provost, 1993). The responses were made using the buttons of a PsyScope response box. The stimulus used was a blue circle (4.5 cm in diameter) presented in the center of the computer screen. Post-response feedback took the form of a picture of a clown presented for 2 s in

the center of the screen, who was either smiling (correct responses) or frowning (incorrect ones).

Procedure

The experimental session consisted of three successive phases: pre-training, training, and testing. For each phase, the experimenter instructed the participants to use the total signal duration as the judgment criterion and to ignore the number of stimuli. In the pre-training phase, the participants were shown two standard sequences of stimuli, each presented five times in alternation. As in Meck and Church (1983) experiment, the first standard sequence was presented for 2 s and consisted of 2 stimuli (“short-few” standard), and the second standard sequence for 8 s with 8 stimuli (“long-many” standard). Each stimulus was displayed for 1 s. During training, the participants were trained to press one button after the “short-few” standard and another one after the “long-many” standard. The button-press order was counterbalanced. Each participant was given successive blocks of 8 trials, with each standard having a 50% probability of appearance on each trial. The intertrial interval was randomly chosen between 1 and 3 s. A correct response resulted in the appearance of the smiling clown and an incorrect one in that of the frowning clown. Training terminated when the participant made no error during a block of 8 trials.

After a successful training block, the participant performed the test phase. This maintained the training conditions, except that the feedback was discontinued. The experimenter said “It’s the same game, but now the clown isn’t here to tell you whether or not you’ve played well.” Furthermore, the participants were presented with two types of comparison sequences. The first held the number of stimuli constant at 4 while varying the total duration of the sequence (2, 3, 4, 5, 6, 7, and 8 s) (“time-varying” dimension). The second held the sequence duration constant at 4 s while varying the number of stimuli (2, 3, 4, 5, 6, 7, and 8 stimuli) (“number-varying” dimension). Thus, each participant saw 8 blocks of 14 trials, 7 for each varying dimension. The 14 trials within each block were presented in a random order.

In each age group, the participants were randomly assigned to a non-counting or a counting group. In the non-counting group, the participants were required to produce aloud repetitive speech as fast as possible (i.e., “blabla”), in order to suppress vocal or subvocal counting (see Baddeley, 1997). The experimenter instructed the participants not to count and monitored the continuity of this verbal activity. In the counting group, they were instructed to count aloud, and to adopt the counting rhythm with which they felt comfortable.

Results

Table 1 shows the number of participants in each group requiring different numbers of training blocks to fulfil the learning criterion (8 consecutive correct responses). In the non-counting group, the required number of blocks differed significantly with age, Kruskal–Wallis test, $X^2(2) = 10.81$, $p = .004$. The between-age comparisons showed that the 5-year-olds required more training blocks than the 8-year-olds, Mann–Whitney U test, $U = 70$, $p = .03$, or the adults, $U = 72$,

Table 1

Number of participants in each age group requiring different numbers of training blocks to fulfil the learning criterion (8 consecutive correct responses) in the training phase for the temporal bisection task in the non-counting and the counting group

	Number of blocks		
	1	2	3
Five-year-olds			
Non-counting	8	4	1
Counting	14	2	0
Eight-year-olds			
Non-counting	15	1	0
Counting	16	0	0
Adults			
Non-counting	18	0	0
Counting	18	0	0

$p = .005$, whereas the 8-year-olds required the same number of blocks as the adults, $U = 135$, $p > .05$. In the counting group, there was no significant effect of age, $\chi^2(2) = 4.33$, $p = .12$. All the participants needed at most 1 or 2 training blocks. Thus, learning the standard duration of a sequence of stimuli took longer in the 5-year-olds than in the older participants only when they did not count.

Fig. 1 shows the mean proportion of “long” responses plotted against stimulus as a function of the varying dimension (time or number) in the 5-year-olds (upper panel), the 8-year-olds (center panel), and the adults (lower panel), for the non-counting (left panels) and counting groups (right panels). An initial overall analysis of variance was run on the slope value of individual psychophysical functions¹ calculated by means of a linear regression, with age and counting as between-subject factors, and varying dimension as within-subject factor.² This ANOVA revealed a main effect of varying dimension, $F(1, 91) = 248.85$, $p = .0001$, but no effect of counting or age. However, there was a significant interaction between counting and varying dimension, $F(1, 91) = 26.17$, $p = .0001$, and between age and varying dimension, $F(2, 91) = 19.56$, $p = .0001$. The other interactions were not significant.

The first significant interaction indicated that the bisection slopes for the time-varying dimension were steeper in the counting than in the non-counting group, $F(1, 95) = 18.24$, $p = .0001$. Conversely, the bisection slopes for the number-varying dimension were flatter in the counting than in the non-counting group, $F(1, 95) = 4.86$, $p = .03$. Thus, a counting strategy improved temporal discrimination and reduced number interference in a temporal bisection task for all age groups.

¹ Analyses of variance were also run on the mean proportion of long responses. Because the results of these ANOVAs were similar to those found by the ANOVA on the individual slope values, only the results of this latter are reported.

² Because initial analyses revealed neither significant main effects nor interactions involving the sex and the button order factors, these factors were not included in the statistical analyses.

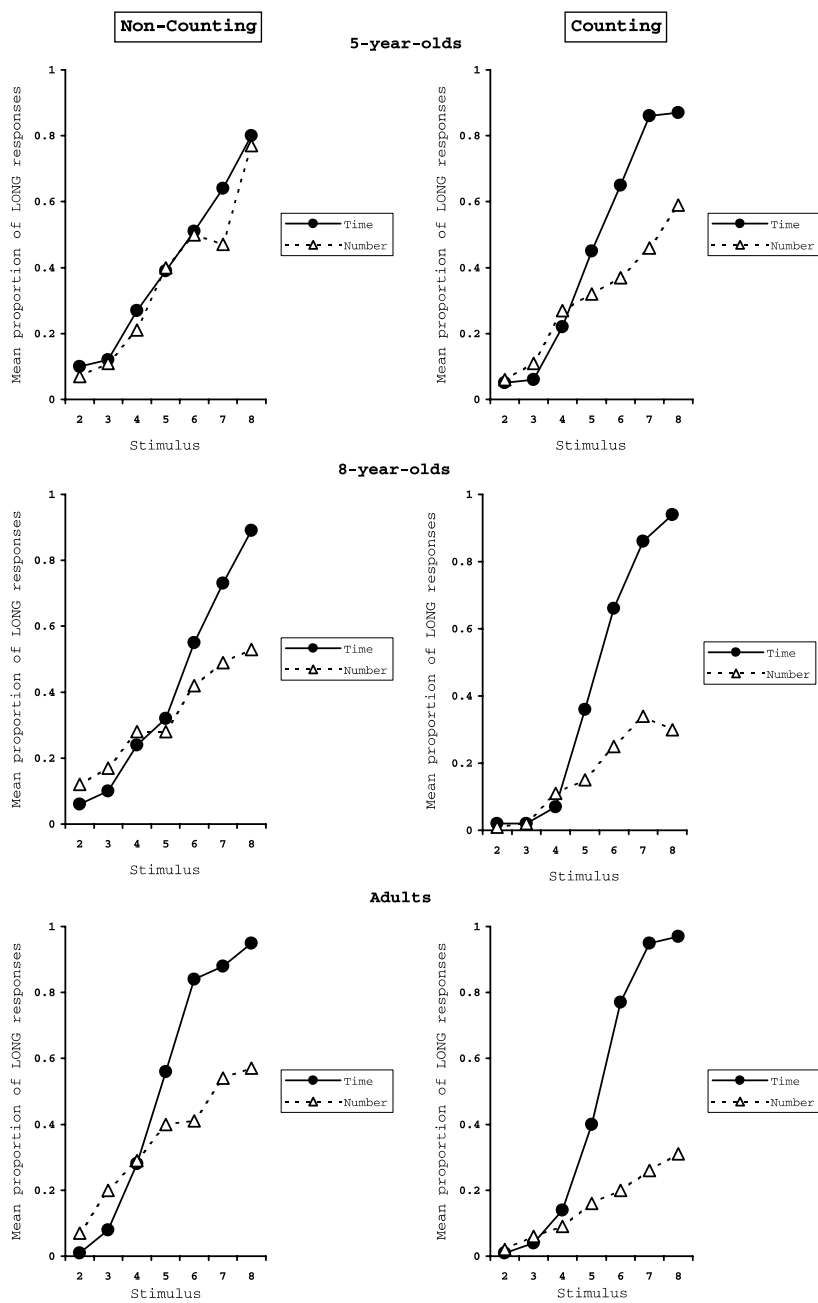


Fig. 1. In the temporal bisection task, proportion of “long” responses plotted against comparison stimulus for the time-varying dimension (time) and the number-varying dimension (number) in the non-counting (left panels) and the counting condition (right panels). Upper panels, data from the 5-year-olds; center panels, data from the 8-year-olds; lowest panels, data from the adults.

The second significant interaction, involving age, indicated that the bisection slope for the time-varying dimension increased with age, $F(2, 96) = 17.69$, $p = .0001$, whereas the bisection slope for the “number-varying” dimension decreased with age, $F(2, 96) = 4.10$, $p = .02$. For the time-varying dimension, the slope value was indeed lower for the 5-year-olds than for the 8-year-olds, and for the 8-year-olds than for the adults (Scheffé post hoc tests, all $p < .05$). Conversely, for the number-varying dimension, the slope value was higher for the 5-year-olds than for the 8-year-olds or the adults ($p < .05$), no difference being observed between the two latter age groups ($p > .05$). However, to provide a clearer picture of the results obtained in the different age groups, we ran an ANOVA on the slope values for each age group separately.

For the 5-year-olds, there was no effect of counting, but a significant effect of varying dimension, $F(1, 27) = 21.08$, $p = .0001$, as well as a significant counting $\times$ varying dimension interaction, $F(1, 27) = 19.50$, $p = .0001$. Indeed, the difference in the slope values between the time- and the number-varying dimension was not significant in the non-counting group, $F(1, 12) = 0.02$, $p = .91$, but significant in the counting group, $F(1, 15) = 42.96$, $p = .0001$. However, even in the counting group, number interfered with temporal performance, as indicated by the proportion of “long” responses that increased with the number of stimuli.

For the 8-year-olds and the adults, there was no effect of counting, but a significant effect of varying dimension ($F(1, 30) = 64.62$, $F(1, 34) = 296.83$, respectively, all $p = .001$), and a significant or a nearly significant interaction between counting and varying dimension ($F(1, 30) = 3.8$, $p = .06$; $F(1, 34) = 10.27$, $p = .003$). Further analysis showed that the effect of the varying-dimension was always significant, both in the non-counting group (8-year-olds, $F(1, 15) = 14.47$, $p = .002$; adults, $F(1, 17) = 126.03$, $p = .001$) and the counting group (8-year-olds, $F(1, 15) = 69.37$; adults, $F(1, 17) = 171.155$, all $p = .0001$). Thus, the 8-year-olds and the adults produced flatter psychophysical functions for the number-varying dimension than for the time-varying dimension in each counting group. Nevertheless, there was still a slight interference of number on temporal performance. Indeed, the proportion of long responses was higher for the comparison sequences with the highest number of stimuli than for those with the smallest number of stimuli (Wilcoxon-test, non-counting group: 8-year-olds, $Z = 2.79$; adults $Z = 3.51$; counting group: 8-year-olds, $Z = 3.20$; adults, $Z = 3.51$, all $p < .005$).

Discussion

Experiment 1 showed that in a temporal bisection task the 5-year-olds were just as able as the 8-year-olds or the adults to discriminate the duration of a sequence of stimuli, although their temporal sensitivity was lower, as shown by the age-related increase in the bisection slope for the time-varying dimension. However, unlike the 8-year-olds or the adults, the 5-year-olds did not succeed in processing time and number independently in a sequence of stimuli without using a verbal counting strategy. Indeed, in the non-counting group, their bisection functions for the time- and the number-varying dimension were similar and superimposed well.

Several studies have shown that young children act inappropriately in a given situation despite knowing what to do. For instance, Zelazo, Frye, and Rapus (1996) showed that although young children were told to use a new rule, and expressed their understanding of this rule, they failed to employ it. They persisted in using the previous rules that were most strongly associated with the task. According to these authors, this failure cannot be explained by young children's lack of understanding of the relevant rule because they represent it, but rather to their inability to resist interference from external stimuli and to inhibit prepotent response tendencies. Indeed, as reported in the introduction, there is ample evidence to show that selective attentional capacities are limited in young children (for a review see Dempster & Brainerd, 1995). The effect of number interference with temporal bisection performance in the 5-year-olds can therefore be explained in terms of their reduced attentional capacities which did not allow them to actively suppress the automatic processing of number in a sequence of events.

Our findings showed that from the age of 8 years, children were more resistant to number interference in a temporal bisection task. Indeed, in the non-counting group, they produced steeper bisection slopes for the time-varying than for the number-varying dimension. However, although reduced, number interference on temporal discrimination persisted in the 8-year-olds and the adults, as indicated by the slight increase in the proportion of long responses when the number of stimuli increased. As shown by our data, a verbal counting strategy reduced number interference, but it did not eliminate it. It was therefore particularly difficult to prevent the processing of number in a sequence of stimuli while attending to its duration.

Given these results, we raised the following question: Does time interfere with the processing of number in a bisection task as number does with the processing of time? The aim of the following experiment was thus to investigate, in the same three age groups, the effect of the duration of a sequence of stimuli on numerical discrimination in a bisection task, when the participants were instructed and trained to process number and to ignore time.

Experiment 2: Numerical bisection

Participants

The sample consisted of 85 participants: 27 5-year-olds (14 girls and 13 boys; mean age = 5.17 years, $SD = 0.40$), 28 8-year-olds (15 girls and 13 boys; mean age = 8.04 years, $SD = 0.33$), and 31 adults (20 women and 11 men; mean age = 22.63 years, $SD = 2.31$).

Material and procedure

The material and the procedure were the same as in experiment 1, except that the participants were instructed to judge the number of stimuli in the signal sequence and to ignore its total duration. The experimenter presented the "short-few" and

the “long-many” standard sequence by saying: “Look, it’s the movie of circles with a few/many circles: there are only a few/lots of circles.” In each age group, the participants were randomly assigned to a counting (for control) or a non-counting group. In the non-counting group, the participants were instructed not to count and counting was prevented by a repetitive speech activity produced as fast as possible and monitored by the experimenter. Conversely, in the counting group, they were instructed to count the number of stimuli aloud.

Results and discussion

In the numerical bisection task, the number of training blocks required to discriminate the two standard sequences did not vary with age, whatever the counting group tested. Most participants required one (25 5-year-olds, 26 8-year-olds, and 30 adults) or two training blocks (2 5-year-olds, 1 8-year-olds, and 1 adult, all in the non-counting groups). Thus, it was relatively easier for the youngest children to learn to discriminate the number of stimuli in a sequence than the total duration of this sequence.

Fig. 2 shows the mean proportion of “many” responses plotted against stimulus for the number- and the time-varying dimension in the non-counting (left panels) and the counting (right panels) groups. The top panels show the data for the 5-year-olds, the center panels the data for the 8-year-olds and the bottom panels the data for the adults. An examination of this figure suggests that for all three age groups the psychophysical function slopes were steep for the number-varying dimension and totally flat for the time-varying dimension. The overall ANOVA performed on the individual regression slope values with age, counting, and varying dimension as factors supported these suggestions. There was a significant main effect of varying dimension, $F(1, 80) = 248.87$, $p = .0001$, and age, $F(2, 80) = 4.48$, $p = .01$, and a significant varying dimension $\times$ counting, $F(1, 80) = 5.04$, $p = .03$, and varying dimension $\times$ age interaction, $F(2, 80) = 3.24$, $p = .04$. The other effects were not significant.

The first significant interaction can be accounted for by the slope values which were steeper with than without counting for the number-varying dimension, $F(1, 85) = 20.14$, $p = .0001$, and equally flat with and without counting for the time-varying dimension, $F(1, 85) = 0.44$, $p = .51$. The second significant interaction indicated that, for the time-varying dimension, the bisection slopes did not differ with age, $F(2, 85) = 2.32$, $p = .11$. Thus, in a numerical bisection task, by age 5 years, children did not produce more “many” responses when the time increased. In contrast, for the number-varying dimension, there was an effect of age, $F(2, 85) = 13.64$, $p = .0001$, indicating that the numerical sensitivity increased between the ages of 5 and 8 years (Sheffé post hoc tests, $p < .05$).

In addition, similarities and differences between temporal and numerical bisection were examined by calculating the bisection point and the Weber ratio from the psychophysical functions. The bisection point is the stimulus value giving rise to 50% “long-many” responses (point of subjective equality). There are various ways of calculating the bisection point but these generally yield nearly identical results (Wearnden & Ferrara, 1995). Here, we used the regression method first employed by

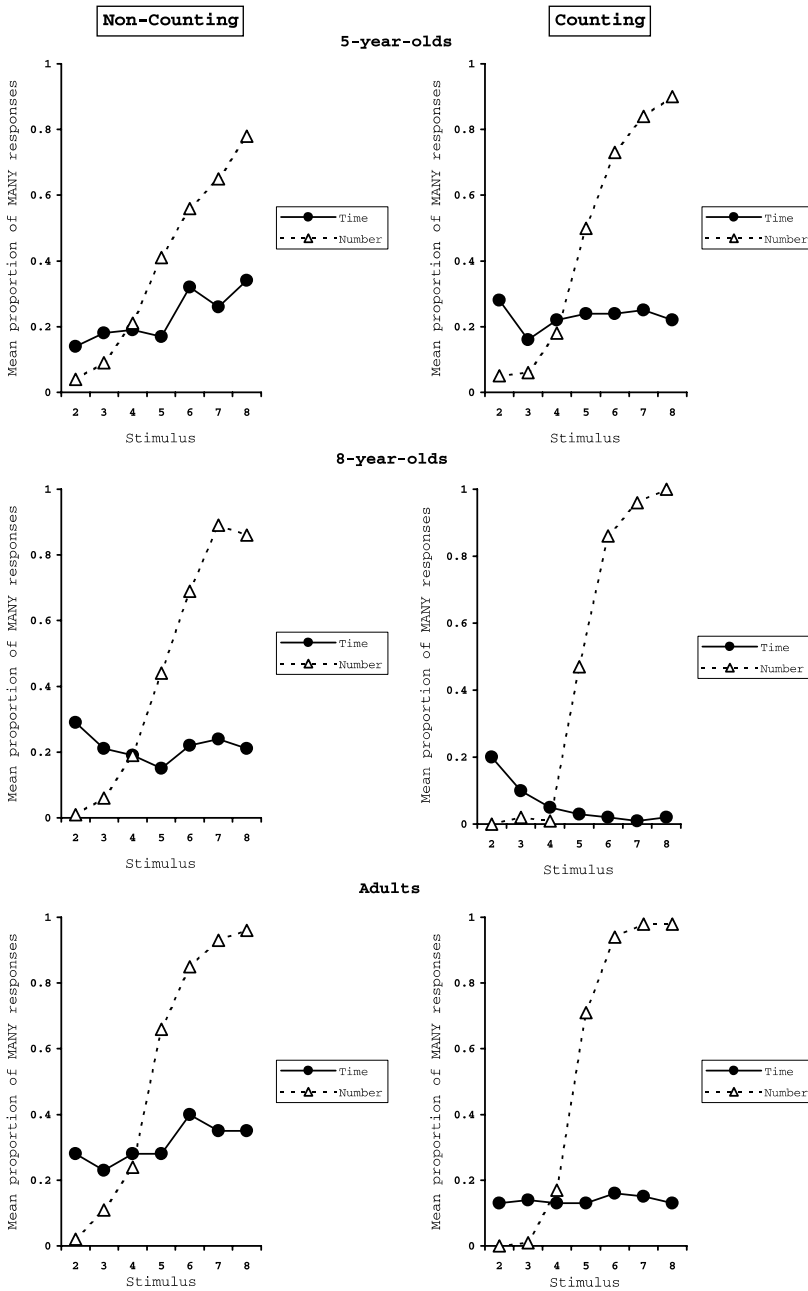


Fig. 2. In the numerical bisection task, proportion of “many” responses plotted against comparison stimulus for the time-varying dimension (time) and the number-varying dimension (number) in the non-counting (left panels) and the counting condition (right panels). Upper panels, data from the 5-year-olds; center panels, data from the 8-year-olds; lowest panels, data from the adults.

Table 2

Bisection points (BP) and Weber ratios (WR) obtained in the 5-year-olds, the 8-year-olds and the adults in the temporal and the numerical bisection task for the non-counting and the counting group

	Five years		Eight years		Adults	
	BP	WR	BP	WR	BP	WR
Temporal bisection						
Non-counting	5.79	0.36	5.60	0.31	4.92	0.28
Counting	5.30	0.29	5.45	0.25	5.16	0.25
Numerical bisection						
Non-counting	5.82	0.32	5.30	0.28	4.79	0.28
Counting	5.20	0.29	5.12	0.24	4.79	0.26

Church and Deluty (1977). The resulting bisection point values are shown in Table 2. For each age group, the bisection points were comparable in temporal and numerical bisection and closer to the arithmetic mean of the two standards (5) than to the geometric mean (4). These results are those classically found in humans (see Allan & Gibbon, 1991; Wearden & Ferrara, 1995).

Table 2 also presents the Weber ratio, a measure of the steepness of the psychophysical function curve, which is in turn an index of sensitivity. This index is calculated from the regression line by, first, determining the difference limen (half the difference between the stimulus value giving rise to 75% “long-many” responses and that giving rise to 25% “long-many” responses), then dividing this difference limen by the bisection point value. A small Weber ratio indicates a steeper psychophysical function and greater temporal or numerical sensitivity. Thus, the Weber ratio was smaller with than without counting in both the temporal and the numerical bisection task. Furthermore, in the non-counting groups, numerical sensitivity was slightly greater than temporal sensitivity in the 5-year-olds and the 8-year-olds, but not in the adults.

Experiment 2 revealed the similarity of performance in a numerical bisection task among the three age groups tested. First, like the 8-year-olds or the adults, the 5-year-olds were able to discriminate sequentially presented numerical quantities, although the numerical sensitivity increased with age. Second, like the older groups, the 5-year-olds were also able to resist time interference when processing the number of a sequence of stimuli. They produced more “many” responses when the number of stimuli increased, but not when the duration increased. Thus, contrary to our assumption, by age 5 years, children were able to process number and time independently in a sequence of stimuli. However, this ability probably depends on the experimental conditions encountered, i.e., the orientation of the attentional focus. As shown by our experiments, the time-number dissociation was easier for the young children when the attentional focus was placed on number (Experiment 2) rather than on time (Experiment 1).

Thus, in a bisection task, number interfered more with time than time did with number. As the extent to which irrelevant information interferes with performance is an indicator of the automatic character of its processing (Logan, 1988), our findings

support the assumption that the processing of number is comparatively less attentionally demanding than that of time. Consequently, because young children have limited attentional capacities, they succeeded better in ignoring temporal information than numerical information and in discriminating number than duration in a sequence of stimuli, as indicated by the fact that the greatest Weber ratio was observed for the numerical than for the temporal bisection.

General discussion

The results of our two experiments provided additional findings showing that timing and counting are two basic abilities shared by human subjects at different ages, although temporal and numerical sensitivity increased with age (e.g., Droit-Volet, 2002; Droit-Volet et al., 2001; Droit-Volet & Wearden, 2001, 2002; Whalen, Gallistel, & Gelman, 1999; Xu & Spelke, 2000). However, they also showed that, in children, numerical sensitivity is greater than temporal sensitivity. Gallistel and Gelman (2000) pointed out a fundamental difference between number and time: number is a “discrete information” whereas time is a “continuous information.” As they explain, the magnitudes representing a countable quantity are generated by a discrete incrementing process which defines the next magnitude just as ordinary counting defines the next value. By contrast, the timing mechanism does not define the next magnitude. As the duration of a timed interval increases, the timing mechanism generates larger magnitudes to represent duration, but it does not select a magnitude that is the next magnitude. The duration is therefore a flow that is evaluated only at the end of an arbitrary interval indicated by a signal, namely the end of the stimulus.

The lower sensitivity for time than for number observed in children can therefore be explained by the greater amount of noise introduced in the representation of the continuous information (time)—presumably young children have difficulty focusing attention until the end of the stimuli to be timed. Indeed, a short period of distraction impairs estimation of time (for a review see Casini & Macar, 1999; Lejeune, 1998), but not estimation of discrete quantities presented sequentially (1 cycle per second). Consequently, the verbal counting strategies improve the accuracy of time judgments (Grondin, Meilleur-Wells, & Lachance, 1999), because they provide both a symbolic label facilitating the memory of duration, and a way of focussing attention on time and maintaining this attentional focus.

The greater sensitivity to number than to duration in a sequence of stimuli also explains in part why number interfered with temporal performance more than time did with numerical performance, and especially in the younger children. As we suggested previously, it was easier for younger children to ignore time than number because the processing of time requires more attention than that of number. Conversely, it was difficult for younger children to ignore number because the processing of number is more automatic and because the inhibition processes become increasingly efficiency with age (Bjorklund & Harnishfeger, 1995). Thus children’s performance in a bisection task with a sequence of stimuli depends on the develop-

ment of attentional processes of both selection and inhibition. This raises a new question for future research: Is temporal and numerical information in a sequence of stimuli selected early by a filter mechanism or/and later through the inhibition of information automatically entering working memory?

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